

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285932

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

HAYASHIGUCHI; Masashi

SEMICONDUCTOR DEVICE AND ELECTRIC POWER CONVERSION UNIT

Abstract

The semiconductor device includes a semiconductor element, a substrate supporting the semiconductor element, a sealing resin covering the semiconductor element and a part of the substrate, and a heat dissipation member. The substrate includes an obverse surface facing a first side in a thickness direction, and a reverse surface facing a second side in the thickness direction and exposed from the sealing resin. The semiconductor element is mounted on the obverse surface. The heat dissipation member is disposed on the substrate on the second side in the thickness direction. The heat dissipation member includes first bases located on the first side in the thickness direction, and first upright portions extending from the first bases to the second side in the thickness direction. The substrate includes first recesses recessed from the reverse surface to the first side in the thickness direction. The first bases are housed in the first recesses.

Inventors: HAYASHIGUCHI; Masashi (Kyoto-shi, JP)

Applicant: Rohm Co., Ltd. (Kyoto-shi, JP)

Family ID: 91323652

Appl. No.: 19/214644

Filed: May 21, 2025

Foreign Application Priority Data

JP 2022-188959

Nov. 28, 2022

Related U.S. Application Data

parent WO continuation PCT/JP2023/041077 20231115 PENDING child US 19214644

Publication Classification

Int. Cl.: H01L23/367 (20060101); **H01L23/00** (20060101); **H01L23/13** (20060101); **H01L23/31** (20060101); **H01L23/373** (20060101); **H01L23/473** (20060101); **H01L23/498** (20060101); **H01L23/538** (20060101); **H01L25/07** (20060101); **H10D80/20** (20250101)

U.S. Cl.:

CPC H01L23/3677 (20130101); **H01L23/13** (20130101); **H01L23/3736** (20130101); **H01L23/473** (20130101); **H01L23/5383** (20130101); **H01L23/5385** (20130101); H01L23/3121 (20130101); H01L23/3735 (20130101); H01L23/49811 (20130101); H01L24/37 (20130101); H01L24/40 (20130101); H01L24/41 (20130101); H01L24/48 (20130101); H01L24/73 (20130101); H01L25/072 (20130101); H01L2224/37147 (20130101); H01L2224/40155 (20130101); H01L2224/4103 (20130101); H01L2224/41051 (20130101); H01L2224/48155 (20130101); H01L2224/73221 (20130101); H01L2924/13091 (20130101); H10D80/251 (20250101)

Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a semiconductor device and an electric power conversion unit.

BACKGROUND ART

[0002] Semiconductor devices provided with power switching elements, such as metal oxide semiconductor field effect transistors (MOSFETs) and insulated gate bipolar transistors (IGBTs), have been conventionally known. Such semiconductor devices are mounted in various electronic devices, ranging from industrial devices to home appliances and information terminals, or even to vehicle-mount devices. JP-A-2021-190505 discloses a conventional semiconductor device (power module). The semiconductor device disclosed in JP-A-2021-190505 includes a semiconductor element and a support substrate. The semiconductor element is an IGBT made of silicon (Si), for example. The support substrate supports the semiconductor element. The support substrate includes an insulating base and conductor layers stacked on the respective surfaces of the base. The base is made of ceramic, for example. Each of the conductor layers is made of copper (Cu), for example, and the semiconductor element is bonded to one of the conductor layers. The semiconductor element is covered with a sealing resin, for example.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. 1 is a perspective view showing a semiconductor device according to a first embodiment of the present disclosure.

[0004] FIG. 2 is a partial perspective view showing the semiconductor device according to the first embodiment of the present disclosure.

[0005] FIG. 3 is a partial perspective view showing the semiconductor device according to the first embodiment of the present disclosure.

[0006] FIG. 4 is a plan view showing the semiconductor device according to the first embodiment of the present disclosure.

[0007] FIG. 5 is a partial plan view showing the semiconductor device according to the first embodiment of the present disclosure.

[0008] FIG. 6 is a partial side view showing the semiconductor device according to the first

embodiment of the present disclosure.

[0009] FIG. 7 is a partially enlarged plan view showing the semiconductor device according to the first embodiment of the present disclosure.

[0010] FIG. 8 is a partial plan view showing the semiconductor device according to the first embodiment of the present disclosure.

[0011] FIG. 9 is a partial plan view showing the semiconductor device according to the first embodiment of the present disclosure.

[0012] FIG. 10 is a side view showing the semiconductor device according to the first embodiment of the present disclosure.

[0013] FIG. 11 is a bottom view showing the semiconductor device according to the first embodiment of the present disclosure.

[0014] FIG. 12 is a partial bottom view showing the semiconductor device according to the first embodiment of the present disclosure.

[0015] FIG. 13 is a cross-sectional view along line XIII-XIII in FIG. 5.

[0016] FIG. 14 is a cross-sectional view along line XIV-XIV in FIG. 5.

[0017] FIG. 15 is a partially enlarged cross-sectional view showing the semiconductor device according to the first embodiment of the present disclosure.

[0018] FIG. 16 is a partially enlarged cross-sectional view showing the semiconductor device according to the first embodiment of the present disclosure.

[0019] FIG. 17 is a cross-sectional view along line XVII-XVII in FIG. 5.

[0020] FIG. 18 is a cross-sectional view along line XVIII-XVIII in FIG. 5.

[0021] FIG. 19 is a cross-sectional view along line XIX-XIX in FIG. 5.

[0022] FIG. 20 is a cross-sectional view along line XX-XX in FIG. 5.

[0023] FIG. 21 is a cross-sectional view along line XXI-XXI in FIG. 5.

[0024] FIG. 22 is a partially enlarged cross-sectional view along line XXII-XXII in FIG. 11.

[0025] FIG. 23 is a cross-sectional view showing an electric power conversion unit according to the first embodiment of the present disclosure.

[0026] FIG. 24 is a cross-sectional view showing the electric power conversion unit according to the first embodiment of the present disclosure.

[0027] FIG. 25 is a bottom view showing a semiconductor device according to a second embodiment of the present disclosure.

[0028] FIG. 26 is a bottom view showing the semiconductor device according to the second embodiment of the present disclosure.

[0029] FIG. 27 is a partially enlarged cross-sectional view along line XXVI-XXVI in FIG. 25.

[0030] FIG. 28 is a side view showing a semiconductor device according to a third embodiment of the present disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

[0031] The following describes preferred embodiments of the present disclosure in detail with reference to the drawings.

[0032] The terms such as “first”, “second” and “third” in the present disclosure are used merely for identification, and are not intended to impose orders on the elements accompanied with these terms.

[0033] In the present disclosure, the phrases “an object A is formed in an object B” and “an object A is formed on an object B” include, unless otherwise specified, “an object A is formed directly in/on an object B” and “an object A is formed in/on an object B with another object interposed between the object A and the object B”. Similarly, the phrases “an object A is disposed in an object B” and “an object A is disposed on an object B” include, unless otherwise specified, “an object A is disposed directly in/on an object B” and “an object A is disposed in/on an object B with another object interposed between the object A and the object B”. Similarly, the phrase “an object A is located on an object B” includes, unless otherwise specified, “an object A is located on an object B

in contact with the object B” and “an object A is located on an object B with another object interposed between the object A and the object B”. Further, the phrase “an object A overlaps with an object B as viewed in a certain direction” includes, unless otherwise specified, “an object A overlaps with the entirety of an object B” and “an object A overlaps with a part of an object B”. Further, the phrase “a plane A faces (a first side or a second side) in a direction B” is not limited to the case where the angle of the plane A with respect to the direction B is 90°, but also includes the case where the plane A is inclined to the direction B.

First Embodiment

[0034] FIGS. **1** to **24** show a semiconductor device and an electric power conversion unit according to a first embodiment of the present disclosure. A semiconductor device **A1** of the present embodiment includes a plurality of first semiconductor elements **10A**, a plurality of second semiconductor elements **10B**, a first heat dissipation member **2A**, a first substrate **3A**, a first terminal **41**, a second terminal **42**, a plurality of third terminals **43**, a fourth terminal **44**, a plurality of control terminals **45**, a control terminal support **48**, a first conductive member **5**, a second conductive member **6**, and a sealing resin **8**.

[0035] FIG. **1** is a perspective view showing the semiconductor device **A1**. FIG. **2** is a partial perspective view showing the semiconductor device **A1**. FIG. **3** is a partial perspective view showing the semiconductor device **A1**. FIG. **4** is a plan view showing the semiconductor device **A1**. FIG. **5** is a partial plan view showing the semiconductor device **A1**. FIG. **6** is a partial side view showing the semiconductor device **A1**. FIG. **7** is a partially enlarged plan view showing the semiconductor device **A1**. FIG. **8** is a partial plan view showing the semiconductor device **A1**. FIG. **9** is a partial plan view showing the semiconductor device **A1**. FIG. **10** is a side view showing the semiconductor device **A1**. FIG. **11** is a bottom view showing the semiconductor device **A1**. FIG. **12** is a partial bottom view showing the semiconductor device **A1**. FIG. **13** is a cross-sectional view along line XIII-XIII in FIG. **5**. FIG. **14** is a cross-sectional view along line XIV-XIV in FIG. **5**. FIG. **15** is a partially enlarged cross-sectional view showing the semiconductor device **A1**. FIG. **16** is a partially enlarged cross-sectional view showing the semiconductor device **A1**. FIG. **17** is a cross-sectional view along line XVII-XVII in FIG. **5**. FIG. **18** is a cross-sectional view along line XVIII-XVIII in FIG. **5**. FIG. **19** is a cross-sectional view along line XIX-XIX in FIG. **5**. FIG. **20** is a cross-sectional view along line XX-XX in FIG. **5**. FIG. **21** is a cross-sectional view along line XXI-XXI in FIG. **5**. FIG. **22** is a partially enlarged cross-sectional view along line XXII-XXII in FIG. **11**. FIGS. **23** and **24** are cross-sectional views showing an electric power conversion unit **B1**. [0036] In these figures, one side in a first direction *x* is referred to as an *x1* side in the first direction *x*, and the other side in the first direction *x* is referred to as an *x2* side in the first direction *x*. The same applies to a second direction *y* and a thickness direction *z*.

[0037] First semiconductor elements **10A** and second semiconductor elements **10B**:

[0038] Each of the first semiconductor elements **10A** and the second semiconductor elements **10B** is an electronic component that forms the functional core of the semiconductor device **A1**. The first semiconductor elements **10A** and the second semiconductor elements **10B** are made of a semiconductor material mainly containing silicon carbide (SiC), for example. The semiconductor material is not limited to SiC, and may be silicon (Si), gallium nitride (GaN), or diamond (C). Each of the first semiconductor elements **10A** and the second semiconductor elements **10B** is a power semiconductor chip having a switching function such as a metal oxide semiconductor field effect transistor (MOSFET). The first semiconductor elements **10A** and the second semiconductor elements **10B** are MOSFETs in the present embodiment, but may be other transistors such as insulated gate bipolar transistors (IGBTs) in other examples. The first semiconductor elements **10A** and the second semiconductor elements **10B** are identical to each other. The first semiconductor elements **10A** and the second semiconductor elements **10B** are n-channel MOSFETs, but may be p-channel MOSFETs instead.

[0039] As shown in FIGS. **15** and **16**, each of the first semiconductor elements **10A** and the second

semiconductor elements **10B** has an element obverse surface **101** and an element reverse surface **102**. In each of the first semiconductor elements **10A** and the second semiconductor elements **10B**, the element obverse surface **101** and the element reverse surface **102** are spaced apart from each other in the thickness direction z . The element obverse surface **101** faces a $z1$ side in the thickness direction z , and the element reverse surface **102** faces a $z2$ side in the thickness direction z .

[0040] In the present embodiment, the semiconductor device **A1** includes four first semiconductor elements **10A** and four second semiconductor elements **10B**. However, the number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** are not limited to the present example, and can be changed appropriately according to the performance required for the semiconductor device **A1**. In the example shown in FIGS. **8** and **9**, four first semiconductor elements **10A** and four second semiconductor elements **10B** are arranged. Each of the number of first semiconductor elements **10A** and the number of second semiconductor elements **10B** may be two or three, or may be five or greater. The number of first semiconductor elements **10A** may be the same as or different from the number of second semiconductor elements **10B**. The respective numbers of first semiconductor elements **10A** and second semiconductor elements **10B** are determined according to the current capacity handled by the semiconductor device **A1**.

[0041] The semiconductor device **A1** is configured as a half-bridge switching circuit, for example. In this case, the first semiconductor elements **10A** form an upper arm circuit of the semiconductor device **A1**, and the second semiconductor elements **10B** form a lower arm circuit. In the upper arm circuit, the first semiconductor elements **10A** are connected in parallel. In the lower arm circuit, the second semiconductor elements **10B** are also connected in parallel. Each first semiconductor element **10A** is connected in series to a second semiconductor element **10B** to form a bridge layer.

[0042] As shown in FIGS. **8**, **9**, and **20** in particular, the first semiconductor elements **10A** are mounted on a below-described first conductive portion **321** of the first substrate **3A**. In the example shown in FIGS. **8** and **9**, the first semiconductor elements **10A** are aligned in the second direction y and spaced apart from each other. Each of the first semiconductor elements **10A** is electrically bonded to the first conductive portion **321** via a first conductive bonding member **19A**. Each of the first semiconductor elements **10A** is bonded to the first conductive portion **321**, such that the element reverse surface **102** faces the first conductive portion **321**. Unlike the present embodiment, the first semiconductor elements **10A** may be mounted on a metal member different from a part of a substrate such as a DBC substrate. In this case, the metal member corresponds to the first conductive portion in the present disclosure. The metal member may be supported by the first conductive portion **321**, for example.

[0043] As shown in FIGS. **8**, **9**, and **19** in particular, the second semiconductor elements **10B** are mounted on a below-described second conductive portion **322** of the first substrate **3A**. In the example shown in FIGS. **8** and **9**, the second semiconductor elements **10B** are aligned in the second direction y and spaced apart from each other. Each of the second semiconductor elements **10B** is electrically bonded to the second conductive portion **322** via a second conductive bonding member **19B**. Each of the second semiconductor elements **10B** is bonded to the second conductive portion **322**, such that the element reverse surface **102** faces the second conductive portion **322**. As can be understood from FIG. **9**, the first semiconductor elements **10A** and the second semiconductor elements **10B** overlap with each other as viewed in the first direction x , but this overlap is not necessary. Unlike the present embodiment, the second semiconductor elements **10B** may be mounted on a metal member different from a part of a substrate such as a DBC substrate. In this case, the metal member corresponds to the second conductive portion in the present disclosure. The metal member may be supported by the second conductive portion **322**, for example.

[0044] Each of the first semiconductor elements **10A** and the second semiconductor elements **10B** includes a first obverse-surface electrode **11**, a second obverse-surface electrode **12**, a third obverse-surface electrode **13**, and a reverse-surface electrode **15**. The description given below of the configurations of the first obverse-surface electrode **11**, the second obverse-surface electrode

12, the third obverse-surface electrode **13**, and the reverse-surface electrode **15** is common to all the first semiconductor elements **10A** and the second semiconductor elements **10B**. The first obverse-surface electrode **11**, the second obverse-surface electrode **12**, and the third obverse-surface electrode **13** are disposed on the element obverse surface **101**. The first obverse-surface electrode **11**, the second obverse-surface electrode **12**, and the third obverse-surface electrode **13** are insulated by a non-illustrated insulating film. The reverse-surface electrode **15** is disposed on the element reverse surface **102**.

[0045] The first obverse-surface electrode **11** is a gate electrode, for example, and receives a drive signal (e.g., gate voltage) inputted to drive the first semiconductor element **10A** (the second semiconductor element **10B**). The second obverse-surface electrode **12** of the first semiconductor element **10A** (the second semiconductor element **10B**) is a source electrode, for example, and conducts a source current. The second obverse-surface electrode **12** of the present embodiment includes a gate finger **121**. The gate finger **121** is a linear insulator that extends in the first direction *x*, for example, and divides the second obverse-surface electrode **12** into two regions in the second direction *y*. The third obverse-surface electrode **13** is a source sense electrode, for example, and conducts the source current. The reverse-surface electrode **15** is a drain electrode, for example, and conducts a drain current. The reverse-surface electrode **15** covers the entirety (or substantially the entirety) of the element reverse surface **102**. The reverse-surface electrode **15** is formed by silver (Ag) plating, for example.

[0046] Each first semiconductor element **10A** (each second semiconductor element **10B**) switches between a conducting state and a non-conducting state in response to a drive signal (gate voltage) inputted to the first obverse-surface electrode **11** (the gate electrode). In the conducting state, a current flows from the reverse-surface electrode **15** (the drain electrode) to the second obverse-surface electrode **12** (the source electrode). In the non-conducting state, the current does not flow. In short, each first semiconductor element **10A** (each second semiconductor element **10B**) performs a switching operation. With the switching functions of the first semiconductor elements **10A** and the second semiconductor elements **10B**, the semiconductor device **A1** converts the DC voltage inputted between the fourth terminal **44** and each of the first terminal **41** and the second terminal **42** into AC voltage, for example, and outputs the AC voltage from the third terminals **43**.

[0047] As shown in FIGS. 5, 8, and 9 in particular, the semiconductor device **A1** includes thermistors **17**. The thermistors **17** are used as temperature detection sensors. Note that the semiconductor device **A1** may include a temperature-sensing diode, for example, in addition to the thermistors **17**, or may not include any thermistors **17**.

First Substrate **3A**:

[0048] The first substrate **3A** supports the first semiconductor elements **10A** and the second semiconductor elements **10B**. The specific configuration of the first substrate **3A** is not particularly limited. For example, the first substrate **3A** may be a direct bonded copper (DBC) substrate or an active metal brazing (AMB) substrate. The first substrate **3A** includes a first insulating layer **31A**, a first obverse-surface metal layer **32A**, and a first reverse-surface metal layer **33A**. The first obverse-surface metal layer **32A** includes the first conductive portion **321** and the second conductive portion **322**. The dimension of the first substrate **3A** in the thickness direction *z* is at least 0.4 mm and at most 3.0 mm, for example.

[0049] The first insulating layer **31A** is made of a ceramic material with excellent thermal conductivity, for example. Examples of such a ceramic material include silicon nitride (SiN). The material of the first insulating layer **31A** is not limited to ceramic, and may be an insulating resin sheet, for example. The first insulating layer **31A** is rectangular in plan view, for example. The dimension of the first insulating layer **31A** in the thickness direction *z* is at least 0.05 mm and at most 1.0 mm, for example.

[0050] The first conductive portion **321** supports the first semiconductor elements **10A**, and the second conductive portion **322** supports the second semiconductor elements **10B**. The first

conductive portion **321** and the second conductive portion **322** are formed on the upper surface (the surface facing the z1 side in the thickness direction z) of the first insulating layer **31A**. The constituent material of each of the first conductive portion **321** and the second conductive portion **322** contains copper (Cu), for example. The constituent material may contain aluminum (Al) instead of copper (Cu), for example. The first conductive portion **321** and the second conductive portion **322** are spaced apart from each other in the first direction x. The first conductive portion **321** is located on the x1 side in the first direction x from the second conductive portion **322**. Each of the first conductive portion **321** and the second conductive portion **322** is rectangular in plan view, for example. The first conductive portion **321** and the second conductive portion **322**, together with the first conductive member **5** and the second conductive member **6**, form the paths of a main circuit current that is switched by the first semiconductor elements **10A** and the second semiconductor elements **10B**.

[0051] The first conductive portion **321** has a first obverse surface **301A**. The first obverse surface **301A** is a flat surface facing the z1 side in the thickness direction z. Each of the first semiconductor elements **10A** is bonded to the first obverse surface **301A** of the first conductive portion **321** via a first conductive bonding member **19A**. The second conductive portion **322** has a second obverse surface **301B**. The second obverse surface **301B** is a flat surface facing the z1 side in the thickness direction z. Each of the second semiconductor elements **10B** is bonded to the second obverse surface **301B** of the second conductive portion **322** via a second conductive bonding member **19B**. The constituent material of each of the first conductive bonding members **19A** and the second conductive bonding members **19B** is not particularly limited, and may be solder, metal paste containing a metal such as silver (Ag), or a sintered metal containing a metal such as silver (Ag). The dimension of each of the first conductive portion **321** and the second conductive portion **322** in the thickness direction z may be at least 0.1 mm and at most 1.5 mm, for example.

[0052] The first reverse-surface metal layer **33A** is formed on the lower surface (the surface facing the z2 side in the thickness direction z) of the first insulating layer **31A**. The constituent material of the first reverse-surface metal layer **33A** is the same as that of the first obverse-surface metal layer **32A**. The first reverse-surface metal layer **33A** has a first reverse surface **302A** and a plurality of first recesses **303A**. The first reverse surface **302A** is a flat surface facing the z2 side in the thickness direction z. The first reverse surface **302A** is exposed from the sealing resin **8**. The first reverse-surface metal layer **33A** overlaps with the first conductive portion **321** and the second conductive portion **322** in plan view.

[0053] As shown in FIGS. **12**, **13**, and **22**, the first recesses **303A** are recessed from the first reverse surface **302A** to the z1 side in the thickness direction z. In the present embodiment, the first recesses **303A** are aligned in the first direction x. The first recesses **303A** according to the present embodiment are grooves extending in the second direction y.

First Heat Dissipation Member **2a**:

[0054] As shown in FIGS. **2**, **6**, **10** to **14**, and **17** to **22**, the first heat dissipation member **2A** is disposed on the surface of the first substrate **3A** (the first reverse-surface metal layer **33A**) on the z2 side in the thickness direction z. The first heat dissipation member **2A** has a plurality of first bases **21A**, a plurality of first upright portions **22A**, and a plurality of first connecting portions **23A**. The material of the first heat dissipation member **2A** is not particularly limited. For example, the first heat dissipation member **2A** is made of a metal plate material. The metal plate material may contain a metal such as copper (Cu), aluminum (Al), or stainless steel, or an alloy of these metals.

[0055] The first bases **21A** are located on the z1 side in the thickness direction z. The first bases **21A** are housed in the respective first recesses **303A**. In the first recesses **303A**, the first bases **21A** are bonded to the first reverse-surface metal layer **33A**. The method for bonding the first bases **21A** to the first reverse-surface metal layer **33A** is not particularly limited, and can be selected as appropriate from a welding method such as laser welding, a method using joints such as brazing, or other methods such as ultrasonic bonding or solid-phase diffusion bonding. In the example shown

in FIG. 22, the first bases **21A** are bonded to the first reverse-surface metal layer **33A** by laser bonding. In this case, parts of the first bases **21A** and parts of the first reverse-surface metal layer **33A** are melted and solidified to form weld portions **M**. In the present embodiment, the thickness of each first base **21A** in the thickness direction **z** is smaller than the depth of each first recess **303A** in the thickness direction **z**. The shape of each first base **21A** is not particularly limited. In the present embodiment, each of the first bases **21A** has a strip shape extending in the second direction **y**. Each of the first bases **21A** may be sized and shaped to fit in a first recess **303A**, or may be slightly smaller than a first recess **303A**.

[0056] The first upright portions **22A** are connected to the respective ends of the first bases **21A** in the first direction **x**, and stand in the thickness direction **z**. The first upright portions **22A** protrude beyond the first reverse surface **302A** to the **z2** side in the thickness direction **z**. In the present embodiment, the length of each first upright portion **22A** in the second direction **y** is equal (or substantially equal) to the length of each first base **21A** in the second direction **y**. Each of the first upright portions **22A** has a strip shape extending in the second direction **y**, for example.

[0057] Each of the first connecting portions **23A** connects the **z2**-side ends in the thickness direction **z** of first upright portions **22A** adjacent to each other in the first direction **x**. In the present embodiment, the length of each first connecting portion **23A** in the second direction **y** is equal (or substantially equal) to the length of each first upright portion **22A** in the second direction **y**. The shape of each first connecting portion **23A** as viewed in the thickness direction **z** is not particularly limited, and may be a strip shape in the present embodiment. The shape of each first connecting portion **23A** as viewed in the second direction **y** is not particularly limited, and may be a flat shape along the first direction **x** as illustrated, or may be a dome shape or a ridge shape that bulges to the **z2** side in the thickness direction **z**.

First Terminal **41**, Second Terminal **42**, Third Terminals **43**, and Fourth Terminal **44**:

[0058] The first terminal **41**, the second terminal **42**, the third terminals **43**, and the fourth terminal **44** are made of metal plates. The metal plates may contain copper (Cu) or a copper (Cu) alloy. In the example shown in FIGS. 1 to 5, 8, 9, and 11, the semiconductor device **A1** includes one first terminal **41**, one second terminal **42**, one fourth terminal **44**, and two third terminals **43**, but the respective numbers of these terminals are not particularly limited.

[0059] The first terminal **41**, the second terminal **42**, and the fourth terminal **44** are input terminals for DC voltage that is to be converted. The fourth terminal **44** is a positive electrode (P terminal), and the first terminal **41** and the second terminal **42** are negative electrodes (N terminals). The third terminals **43** are output terminals for the AC voltage resulting from the power conversion by the first semiconductor elements **10A** and the second semiconductor elements **10B**. Each of the first terminal **41**, the second terminal **42**, the third terminals **43**, and the fourth terminal **44** includes a portion covered with the sealing resin **8** and a portion exposed from the sealing resin **8**.

[0060] As shown in FIG. 14, the fourth terminal **44** is electrically bonded to the first conductive portion **321**. The method for electrical bonding is not particularly limited, and may be ultrasonic bonding, laser bonding, welding, or bonding with solder, metal paste or a sintered silver, as appropriate. As shown in FIGS. 8 and 9 in particular, the fourth terminal **44** is located on the **x1** side in the first direction **x** from the first semiconductor elements **10A** and the first conductive portion **321**. The fourth terminal **44** is electrically connected to the first conductive portion **321**, and also to the reverse-surface electrodes **15** (the drain electrodes) of the first semiconductor elements **10A** via the first conductive portion **321**.

[0061] The first terminal **41** and the second terminal **42** are electrically connected to the second conductive member **6**. In the present embodiment, the first terminal **41** and the second conductive member **6** are integrally formed. The first terminal **41** and the second conductive member **6** that are integrally formed have no bonding material or joint, and they may be formed by cutting and bending a single metal plate, for example. In the present embodiment, the second terminal **42** and the second conductive member **6** are also integrally formed. As long as the first terminal **41** and the

second terminal **42** are electrically connected to the second conductive member **6**, they may be separate components unlike the present embodiment, and may include bonding portions bonded to the second conductive member **6**. As shown in FIGS. **5** and **8** in particular, the first terminal **41** and the second terminal **42** are located on the x1 side in the first direction x from the first semiconductor elements **10A** and the first conductive portion **321**. The first terminal **41** and the second terminal **42** are electrically connected to the second conductive member **6**, and also to the second obverse-surface electrodes **12** (the source electrodes) of the respective second semiconductor elements **10B** via the second conductive member **6**.

[0062] As shown in FIGS. **1** to **5**, and **11**, the first terminal **41**, the second terminal **42**, and the fourth terminal **44** of the semiconductor device **A1** protrude from the sealing resin **8** to the x1 side in the first direction x. The first terminal **41**, the second terminal **42**, and the fourth terminal **44** are spaced apart from each other. The first terminal **41** and the second terminal **42** are located opposite to each other across the fourth terminal **44** in the second direction y. The first terminal **41** is located on the y1 side in the second direction y from the fourth terminal **44**, and the second terminal **42** is located on the y2 side in the second direction y from the fourth terminal **44**. The first terminal **41**, the second terminal **42**, and the fourth terminal **44** overlap with each other as viewed in the second direction y.

[0063] As can be seen from FIGS. **8**, **9**, and **13**, the two third terminals **43** are electrically bonded to the second conductive portion **322**. The method for electrical bonding is not particularly limited, and may be ultrasonic bonding, laser bonding, welding, or bonding with solder, metal paste or a sintered silver, as appropriate. As shown in FIG. **8** in particular, the two third terminals **43** are located on the x2 side in the first direction x from the second semiconductor elements **10B** and the second conductive portion **322**. The third terminals **43** are electrically connected to the second conductive portion **322**, and also to the reverse-surface electrodes **15** (the drain electrodes) of the second semiconductor elements **10B** via the second conductive portion **322**. Note that the number of third terminals **43** is not limited to two, and may be one or greater than two. If one third terminal **43** is provided, it is preferable that the third terminal **43** be connected to the central portion of the second conductive portion **322** in the second direction y.

[0064] The control terminals **45** are pin-like terminals for controlling the first semiconductor elements **10A** and the second semiconductor elements **10B**. The control terminals **45** include a plurality of first control terminals **46A** to **46E** and a plurality of second control terminals **47A** to **47D**. The first control terminals **46A** to **46E** are used to control the first semiconductor elements **10A**, for example. The second control terminals **47A** to **47D** are used to control the second semiconductor elements **10B**, for example.

First Control Terminals **46a** to **46e**:

[0065] The first control terminals **46A** to **46E** are spaced apart from each other in the second direction y. As shown in FIGS. **8**, **14**, and **21** in particular, the first control terminals **46A** to **46E** are supported by the first conductive portion **321** via the control terminal support **48** (a first support portion **48A** described below). As shown in FIGS. **5** and **8**, the first control terminals **46A** to **46E** are located between the first semiconductor elements **10A** and the first, second, and fourth terminals **41**, **42**, and **44** in the first direction x.

[0066] The first control terminal **46A** is a terminal (a gate terminal) for receiving input of a drive signal for the first semiconductor elements **10A**. The first control terminal **46A** receives a drive signal (e.g., gate voltage) for driving the first semiconductor elements **10A**.

[0067] The first control terminal **46B** is a terminal (a source sense terminal) for detecting the source signal of the first semiconductor elements **10A**. The voltage (the voltage corresponding to the source current) applied to the second obverse-surface electrodes **12** (the source electrodes) of the first semiconductor elements **10A** is detected at the first control terminal **46B**.

[0068] The first control terminals **46C** and **46D** are electrically connected to a thermistor **17**.

[0069] The first control terminal **46E** is a terminal (a drain sense terminal) for detecting the drain

signal of the first semiconductor elements **10A**. The voltage (the voltage corresponding to the drain current) applied to the reverse-surface electrodes **15** (the drain electrodes) of the first semiconductor elements **10A** is detected at the first control terminal **46E**.

[0070] The second control terminals **47A** to **47D** are spaced apart from each other in the second direction y. As shown in FIGS. **8** and **14** in particular, the second control terminals **47A** to **47D** are supported by the second conductive portion **322** via the control terminal support **48** (a second support portion **48B** described below). As shown in FIGS. **5** and **8**, the second control terminals **47A** to **47D** are located between the second semiconductor elements **10B** and the two third terminals **43** in the first direction x.

[0071] The second control terminal **47A** is a terminal (a gate terminal) for receiving input of a drive signal for the second semiconductor elements **10B**. The second control terminal **47A** receives a drive signal (e.g., gate voltage) for driving the second semiconductor elements **10B**. The second control terminal **47B** is a terminal (a source sense terminal) for detecting the source signal of the second semiconductor elements **10B**. The voltage (the voltage corresponding to the source current) applied to the second obverse-surface electrodes **12** (the source electrodes) of the second semiconductor elements **10B** is detected at the second control terminal **47B**. The second control terminals **47C** and **47D** are electrically connected to a thermistor **17**.

[0072] Each of the control terminals **45** (the first control terminals **46A** to **46E** and the second control terminals **47A** to **47D**) includes a holder **451** and a metal pin **452**.

[0073] The holder **451** is made of a conductive material. As shown in FIGS. **15** and **16**, the holder **451** is bonded to the control terminal support **48** (a first metal layer **482** described below) via a conductive bonding member **459**. The holder **451** includes a tubular portion, an upper flange, and a lower flange. The upper flange is connected to the upper end of the tubular portion, and the lower flange is connected to the lower end of the tubular portion. The metal pin **452** is inserted through at least the upper flange and the tubular portion of the holder **451**. The holder **451** is covered with the sealing resin **8** (a second protrusion **852** described below).

[0074] The metal pin **452** is a rod-like member extending in the thickness direction z. The metal pin **452** is pressed into the holder **451** and supported by the holder **451**. The metal pin **452** is electrically connected to the control terminal support **48** (the first metal layer **482** described below) at least through the holder **451**. When the lower end (the end on the z2 side in the thickness direction z) of the metal pin **452** is in contact with the conductive bonding member **459** within the insertion hole of the holder **451** as in the example shown in FIGS. **15** and **16**, the electrical connection of the metal pin **452** to the control terminal support **48** is established through the conductive bonding member **459**.

Control Terminal Support **48**:

[0075] The control terminal support **48** supports the control terminals **45**. In the thickness direction z, the control terminal support **48** is located between the first and second obverse surfaces **301A** and **301B** and the plurality of control terminals **45**.

[0076] The control terminal support **48** includes a first support portion **48A** and a second support portion **48B**. The first support portion **48A** is disposed on the first conductive portion **321** and supports the first control terminals **46A** to **46E** out of the plurality of control terminals **45**. As shown in FIG. **15**, the first support portion **48A** is bonded to the first conductive portion **321** via a bonding member **49**. The bonding member **49** can be either conductive or insulating, and solder is used in one example. The second support portion **48B** is disposed on the second conductive portion **322** and supports the second control terminals **47A** to **47D** out of the plurality of control terminals **45**. As shown in FIG. **16**, the second support portion **48B** is bonded to the second conductive portion **322** via a bonding member **49**.

[0077] The control terminal support **48** (each of the first support portion **48A** and the second support portion **48B**) may be composed of a direct bonded copper (DBC) substrate, for example. The control terminal support **48** includes a stack of an insulating layer **481**, a first metal layer **482**,

and a second metal layer **483**.

[0078] The insulating layer **481** is made of a ceramic material, for example. The insulating layer **481** is rectangular in plan view, for example.

[0079] As shown in FIGS. **15** and **16** in particular, the first metal layer **482** is formed on the upper surface of the insulating layer **481**. Each of the control terminals **45** stands on the first metal layer **482**. The first metal layer **482** contains copper (Cu) or a copper (Cu) alloy, for example. As shown in FIG. **8** in particular, the first metal layer **482** includes a first region **482A**, a second region **482B**, a third region **482C**, a fourth region **482D**, a fifth region **482E**, and a sixth region **482F**. The first region **482A**, the second region **482B**, the third region **482C**, the fourth region **482D**, the fifth region **482E**, and the sixth region **482F** are spaced apart and insulated from each other.

[0080] A plurality of wires **71** are bonded to the first region **482A**. The wires **71** electrically connect the first region **482A** to the first obverse-surface electrodes **11** (the gate electrodes) of the first semiconductor elements **10A** (the second semiconductor elements **10B**). A plurality of wires **73** are connected to the first region **482A** and the sixth region **482F**. Thus, the sixth region **482F** is electrically connected to the first obverse-surface electrodes **11** (the gate electrodes) of the first semiconductor elements **10A** (the second semiconductor elements **10B**) via the wires **73** and **71**. As shown in FIG. **8**, the first control terminal **46A** is bonded to the sixth region **482F** of the first support portion **48A**, and the second control terminal **47A** is bonded to the sixth region **482F** of the second support portion **48B**.

[0081] A plurality of wires **72** are bonded to the second region **482B**. The wires **72** electrically connect the second region **482B** to the third obverse-surface electrodes **13** (the source sense electrodes) of the first semiconductor elements **10A** (the second semiconductor elements **10B**). As shown in FIG. **8**, the first control terminal **46B** is bonded to the second region **482B** of the first support portion **48A**, and the second control terminal **47B** is bonded to the second region **482B** of the second support portion **48B**.

[0082] A thermistor **17** is bonded to the third region **482C** and the fourth region **482D**. As shown in FIG. **8**, the first control terminals **46C** and **46D** are respectively bonded to the third region **482C** and the fourth region **482D** of the first support portion **48A**. In addition, the second control terminals **47C** and **47D** are respectively bonded to the third region **482C** and the fourth region **482D** of the second support portion **48B**.

[0083] A wire **74** is bonded to the fifth region **482E** of the first support portion **48A**. The wire **74** electrically connects the fifth region **482E** to the first conductive portion **321**. As shown in FIG. **8**, the first control terminal **46E** is bonded to the fifth region **482E** of the first support portion **48A**. The fifth region **482E** of the second support portion **48B** is not electrically connected to any element. The wires **71** to **74** may be bonding wires, for example. The constituent material of the wires **71** to **74** may contain gold (Au), aluminum (Al), or copper (Cu), for example.

[0084] As shown in FIGS. **15** and **16** in particular, the second metal layer **483** is formed on the lower surface of the insulating layer **481**. As shown in FIG. **15**, the second metal layer **483** of the first support portion **48A** is bonded to the first conductive portion **321** via the bonding member **49**. As shown in FIG. **16**, the second metal layer **483** of the second support portion **48B** is bonded to the second conductive portion **322** via the bonding member **49**.

First Conductive Member **5** and Second Conductive Member **6**:

[0085] The first conductive member **5** and the second conductive member **6**, together with the first conductive portion **321** and the second conductive portion **322**, form the paths of the main circuit current that is switched by the first semiconductor elements **10A** and the second semiconductor elements **10B**. The first conductive member **5** and the second conductive member **6** are spaced apart from the first obverse surface **301A** and the second obverse surface **301B** to the z1 side in the thickness direction z. In plan view, the first conductive member **5** and the second conductive member **6** overlap with the first obverse surface **301A** and the second obverse surface **301B**. In the present embodiment, each of the first conductive member **5** and the second conductive member **6** is

made of a metal plate. The metal may contain copper (Cu) or a copper (Cu) alloy, for example. Specifically, the first conductive member **5** and the second conductive member **6** are metal plates having been bent as needed.

[0086] The first conductive member **5** is connected to the second obverse-surface electrodes **12** (the source electrodes) of the first semiconductor elements **10A** and the second conductive portion **322**, and electrically connects the second obverse-surface electrodes **12** of the first semiconductor elements **10A** and the second conductive portion **322**. The first conductive member **5** forms a path of the main circuit current that is switched by the first semiconductor elements **10A**. As shown in FIGS. **7** and **8**, the first conductive member **5** includes a main portion **51**, a plurality of first bonding portions **52**, and a plurality of second bonding portions **53**.

[0087] The main portion **51** is located between the first semiconductor elements **10A** and the second conductive portion **322** in the first direction **x**, and has the shape of a strip extending in the second direction **y** in plan view. The main portion **51** overlaps with both the first conductive portion **321** and the second conductive portion **322** in plan view, and is spaced apart from the first obverse surface **301A** and the second obverse surface **301B** to the **z1** side in the thickness direction **z**. As shown in FIG. **17** in particular, the main portion **51** is located on the **z2** side in the thickness direction **z** from a plurality of third path portions **66** and a fourth path portion **67** of the second conductive member **6** described below, and is closer to the first obverse surface **301A** and the second obverse surface **301B** than the third path portions **66** and the fourth path portion **67**.

[0088] In the present embodiment, the main portion **51** is parallel to the first obverse surface **301A** and the second obverse surface **301B**.

[0089] As shown in FIG. **8** in particular, the main portion **51** extends continuously in the second direction **y** to correspond to a region in which the first semiconductor elements **10A** are positioned. In the present embodiment, the main portion **51** has a plurality of first openings **514** as shown in FIGS. **7**, **8**, and **14** in particular. The first openings **514** may be through-holes extending in the thickness direction **z** (the direction of the plate thickness of the main portion **51**), for example. The first openings **514** are spaced apart from each other in the second direction **y**. The first openings **514** are provided for the respective first semiconductor elements **10A**. In the present embodiment, the main portion **51** has four first openings **514**, each of which corresponds in position in the second direction **y** to one of the plurality of (four) first semiconductor elements **10A**.

[0090] As shown in FIGS. **8** and **14** in particular, the first openings **514** of the present embodiment overlap with the gap between the first conductive portion **321** and the second conductive portion **322** in plan view. The first openings **514** are provided to facilitate the flow of a molten resin material between the upper side (on the **z1** side in the thickness direction **z**) and the lower side (on the **z2** side in the thickness direction **z**) around the main portion **51** (the first conductive member **5**) when the molten resin material is injected in the process of forming the sealing resin **8**.

[0091] As shown in FIG. **8** in particular, the first bonding portions **52** and the second bonding portions **53** are connected to the main portion **51** and disposed to correspond to the first semiconductor elements **10A**. Specifically, the first bonding portions **52** are located on the **x1** side in the first direction **x** from the main portion **51**. The second bonding portions **53** are located on the **x2** side in the first direction **x** from the main portion **51**. As shown in FIG. **15**, each of the first bonding portions **52** is bonded to the second obverse-surface electrode **12** of a corresponding first semiconductor element **10A** via a conductive bonding member **59**. Each of the second bonding portions **53** is bonded to the second conductive portion **322** via a conductive bonding member **59**. The material of the conductive bonding members **59** is not particularly limited, and may be solder, metal paste, or sintered metal. In the present embodiment, each of the first bonding portions **52** includes two regions spaced apart from each other in the second direction **y**. The two regions are bonded to the second obverse-surface electrode **12** of a corresponding first semiconductor element **10A** on the respective sides of the gate finger **121** of the second obverse-surface electrode **12** in the second direction **y**.

[0092] The second conductive member **6** electrically connects the second obverse-surface electrodes **12** (the source electrodes) of the second semiconductor elements **10B** to the first terminal **41** and the second terminal **42**. The second conductive member **6** is integrally formed with the first terminal **41** and the second terminal **42**. The second conductive member **6** forms a path of the main circuit current that is switched by the second semiconductor elements **10B**. As shown in FIGS. **3**, **5** to **7**, **13**, **14**, **17** to **19**, the second conductive member **6** includes a plurality of third bonding portions **61**, a first path portion **64**, a second path portion **65**, a plurality of third path portions **66**, and a fourth path portion **67**. In the illustrated example, the second conductive member **6** includes a first ramp portion **602** and a second ramp portion **603**.

[0093] The third bonding portions **61** are bonded to the respective second semiconductor elements **10B**. Each of the third bonding portions **61** is bonded to the second obverse-surface electrode **12** of a second semiconductor element **10B** via a conductive bonding member **69**. The material of the conductive bonding members **69** is not particularly limited, and may be solder, metal paste, or sintered metal. In the present embodiment, each of the third bonding portions **61** includes two flat sections **611** and two first inclined sections **612**.

[0094] The two flat sections **611** are aligned in the second direction *y*. The two flat sections **611** are spaced apart from each other in the second direction *y*. The shape of each flat section **611** is not particularly limited, and is rectangular in the illustrated example. The two flat sections **611** are bonded to the second obverse-surface electrode **12** of a corresponding second semiconductor element **10B** on the respective sides of the gate finger **121** of the second obverse-surface electrode **12** in the second direction *y*.

[0095] The two first inclined sections **612** are connected to the outer ends of the respective two flat sections **611** in the second direction *y*. In other words, the first inclined section **612** on the *y1* side in the second direction *y* is connected to the *y1*-side end of the flat section **611** located on the *y1* side in the second direction *y*. The first inclined section **612** located on the *y2* side in the second direction *y* is connected to the *y2*-side end of the flat section **611** located on the *y2* side in the second direction *y*. Each of the first inclined sections **612** is inclined toward the *z1* side in the thickness direction *z* with an increasing distance from the flat section **611** in the second direction *y*.

[0096] The first path portion **64** is located between the third bonding portions **61** and the first terminal **41**. In the illustrated example, the first path portion **64** is connected to the first terminal **41** via the first ramp portion **602**. The first path portion **64** overlaps with the first conductive portion **321** in plan view. The first path portion **64** generally extends in the first direction *x*.

[0097] The first path portion **64** includes a first band-shaped section **641** and a first extended section **643**. The first band-shaped section **641** is located on the *x2* side in the first direction *x* from the first terminal **41**, and is parallel to (or substantially parallel to) the first obverse surface **301A**. The first band-shaped section **641** generally extends in the first direction *x*. In the illustrated example, the first band-shaped section **641** has a recess **649**. The recess **649** is a portion of the first band-shaped section **641** that is recessed toward the *y1* side in the second direction *y*. In FIG. **5**, a first conductive portion **321** is visible through the recess **649**.

[0098] The first extended section **643** extends from the *y1*-side end of the first band-shaped section **641** in the second direction *y* toward the *z2* side in the thickness direction *z*. The first extended section **643** is spaced apart from the first conductive portion **321**. In the illustrated example, the first extended section **643** extends in the thickness direction *z* and has a rectangular shape elongated in the first direction *x*. Note that the first path portion **64** may be configured without the first extended section **643**.

[0099] The second path portion **65** is located between the third bonding portions **61** and the second terminal **42**. In the illustrated example, the second path portion **65** is connected to the second terminal **42** via the second ramp portion **603**. The second path portion **65** overlaps with the first conductive portion **321** in plan view. The second path portion **65** generally extends in the first direction *x*.

[0100] The second path portion **65** includes a second band-shaped section **651** and a second extended section **653**. The second band-shaped section **651** is located on the x2 side in the first direction x from the second terminal **42**, and is parallel to (or substantially parallel to) the first obverse surface **301A**. The second band-shaped section **651** generally extends in the first direction x. In the illustrated example, the second band-shaped section **651** has a recess **659**. The recess **659** is a portion of the second band-shaped section **651** that is recessed toward the y2 side in the second direction y. In FIG. 5, a first conductive portion **321** is visible through the recess **659**.

[0101] The second extended section **653** extends from the y2-side end of the second band-shaped section **651** in the second direction y toward the z2 side in the thickness direction z. The second extended section **653** is spaced apart from the first conductive portion **321**. In the illustrated example, the second extended section **653** extends in the thickness direction z and has a rectangular shape that is elongated in the first direction x. Note that the second path portion **65** may be configured without the second extended section **653**.

[0102] The third path portions **66** are individually connected to the third bonding portions **61**. The third path portions **66** extend in the first direction x, and are spaced apart from each other in the second direction y. The number of third path portions **66** is not particularly limited. In the illustrated example, five third path portions **66** are provided. Each of the third path portions **66** is located either between two of the second semiconductor elements **10B** in the second direction y or outside the second semiconductor elements **10B** in the second direction y.

[0103] The two outermost third path portions **66** in the second direction y are formed with recesses **669**. Each of the recesses **669** is recessed from the inner side toward the outer side in the second direction y. In the illustrated example, each of the two outermost third path portions **66** has one recess **669**. In FIG. 5, the second conductive portion **322** is visible through the recesses **669**.

[0104] In the present embodiment, each of the third bonding portions **61** is located between two third path portions **66** adjacent in the second direction y. Each of the third bonding portions **61** has two first inclined sections **612**, one on the y1 side in the second direction y and the other on the y2 side in the second direction y. The first inclined section **612** on the y1 side is connected to one of the two adjacent third path portions **66** that is located on the y1 side in the second direction y. The first inclined section **612** on the y2 side is connected to one of the two adjacent third path portions **66** that is located on the y2 side in the second direction y.

[0105] The fourth path portion **67** is connected to the ends of the respective third path portions **66** on the x1 side in the first direction x. The fourth path portion **67** extends in the second direction y. The fourth path portion **67** is connected to the x2-side end of the first band-shaped section **641** of the first path portion **64** in the first direction x, and to the x2-side end of the second band-shaped section **651** of the second path portion **65** in the first direction x. In the illustrated example, the fourth path portion **67** is connected to the first path portion **64** at the end on the y1 side in the second direction y and to the second path portion **65** at the end on the y2 side in the second direction y.

Sealing Resin **8**:

[0106] The sealing resin **8** covers the first semiconductor elements **10A**, the second semiconductor elements **10B**, the first substrate **3A** (except for the first reverse surface **302A**), a part of each of the first terminal **41**, the second terminal **42**, the third terminals **43**, and the fourth terminal **44**, a part of each of the control terminals **45**, the control terminal support **48**, the first conductive member **5**, the second conductive member **6**, and the wires **71** to **74**. The sealing resin **8** may be made of a black epoxy resin, for example. The sealing resin **8** may be formed by molding, for example. The sealing resin **8** may have a dimension of about 35 mm to 60 mm in the first direction x, a dimension of about 35 mm to 50 mm in the second direction y, and a dimension of about 4 mm to 15 mm in the thickness direction z. These dimensions are measured at the largest portions in the respective directions. The sealing resin **8** has a resin obverse surface **81**, a resin reverse surface **82**, and resin side surfaces **831** to **834**.

[0107] As shown in FIGS. 10, 13, and 19 in particular, the resin obverse surface **81** and the resin reverse surface **82** are spaced apart from each other in the thickness direction *z*. The resin obverse surface **81** faces the *z1* side in the thickness direction *z*, and the resin reverse surface **82** faces the *z2* side in the thickness direction *z*. The control terminals **45** (the first control terminals **46A** to **46E** and the second control terminals **47A** to **47D**) protrude from the resin obverse surface **81**. As shown in FIG. 11, the resin reverse surface **82** has the shape of a frame surrounding the first reverse surface **302A** (the lower surface of the first reverse-surface metal layer **33A**) of the first substrate **3A** in plan view. The first reverse surface **302A** of the first substrate **3A** is exposed from the resin reverse surface **82**, and is flush with the resin reverse surface **82**, for example. The resin side surfaces **831** to **834** are connected to both the resin obverse surface **81** and the resin reverse surface **82**, and are located between the resin obverse surface **81** and the resin reverse surface **82** in the thickness direction *z*. As shown in FIG. 4 in particular, the resin side surface **831** and the resin side surface **832** are spaced apart from each other in the first direction *x*. The resin side surface **831** faces the *x2* side in the first direction *x*, and the resin side surface **832** faces the *x1* side in the first direction *x*. The two third terminals **43** protrude from the resin side surface **831**, and the first terminal **41**, the second terminal **42**, and the fourth terminal **44** protrude from the resin side surface **832**. As shown in FIG. 4 in particular, the resin side surface **833** and the resin side surface **834** are spaced apart from each other in the second direction *y*. The resin side surface **833** faces the *y2* side in the second direction *y*, and the resin side surface **834** faces the *y1* side in the second direction *y*. [0108] As shown in FIG. 4, the resin side surface **832** has a plurality of recesses **832a**. Each of the recesses **832a** is recessed in the first direction *x* in plan view. The recesses **832a** include one formed between the first terminal **41** and the fourth terminal **44**, and one formed between the second terminal **42** and the fourth terminal **44** in plan view. The recesses **832a** are provided to increase the creepage distance along the resin side surface **832** between the first terminal **41** and the fourth terminal **44**, and also to increase the creepage distance along the resin side surface **832** between the second terminal **42** and the fourth terminal **44**.

[0109] As shown in FIGS. 13 and 14 in particular, the sealing resin **8** includes a plurality of first protrusions **851**, a plurality of second protrusions **852**, and a resin cavity **86**.

[0110] The first protrusions **851** protrude from the resin obverse surface **81** in the thickness direction *z*. The first protrusions **851** are located near the four corners of the sealing resin **8** in plan view. Each of the first protrusions **851** has a first protrusion end surface **851a** at its end (the end on the *z1* side in the thickness direction *z*). The first protrusion end surface **851a** of each first protrusion **851** is parallel to (or substantially parallel to) the resin obverse surface **81** and located in the same plane (*x-y* plane). Each of the first protrusions **851** has the shape of a hollow truncated cone with a bottom, for example. The first protrusions **851** serve as spacers when the semiconductor device **A1** is mounted on, for example, a control circuit board of a device that operates with the power generated by the semiconductor device **A1**. Each of the first protrusions **851** has a recess **851b** and an inner wall surface **851c** of the recess **851b**. Each of the first protrusions **851** is columnar, which is preferably a cylindrical column. The recess **851b** has a cylindrical shape, preferably with the inner wall surface **851c** defining a perfect circle in plan view.

[0111] The sealing resin **8** includes a groove **89**. The groove **89** is recessed from the resin reverse surface **82** to the *z1* side in the thickness direction *z*. The groove **89** extends across the resin reverse surface **82** in the second direction *y*. In the illustrated example, the sealing resin **8** has two grooves **89**. The two grooves **89** are spaced apart from each other in the first direction *x*. The first reverse-surface metal layer **33A** (the first reverse surface **302A**) is located between the two grooves **89**.

[0112] The semiconductor device **A1** may be mechanically fastened to the control circuit board or the like by screwing, for example. In such a case, each first protrusion **851** may be formed with an internal thread on the inner wall surface **851c** of the recess **851b**. For example, an insert nut may be inserted into the recess **851b** of each first protrusion **851**.

[0113] As shown in FIG. 14 in particular, the second protrusions **852** protrude from the resin

obverse surface **81** in the thickness direction **z**. The second protrusions **852** overlap with the control terminals **45** in plan view. The metal pin **452** of each control terminal **45** protrudes from a second protrusion **852**. Each of the second protrusions **852** has the shape of a truncated cone. Each of the second protrusions **852** covers the holder **451** and a portion of the metal pin **452** of a control terminal **45**.

[0114] As shown in FIGS. **23** and **24**, the electric power conversion unit **B1** includes the semiconductor device **A1** and a cooling device **9**.

[0115] The cooling device **9** is disposed on the **z2** side in the thickness direction **z** with respect to the semiconductor device **A1**. The cooling device **9** has a housing **91**.

[0116] The housing **91** is a box-shaped member made of metal or resin, for example. The housing **91** houses the first heat dissipation member **2A**. In the present embodiment, the housing **91** is attached to the semiconductor device **A1** via a sealant **919**. The sealant **919** is disposed between an end of the housing **91** and the resin reverse surface **82** of the sealing resin **8**, and maintains the airtightness of the internal space of the housing **91**. In the present embodiment, the housing **91** is formed with a groove **911**. The groove **911** has an annular shape as viewed in the thickness direction **z**, and houses a part of the sealant **919**.

[0117] The housing **91** is filled with a cooling medium **Cm**. The cooling medium **Cm** flows within the housing **91**. In the present embodiment, the cooling device **9** has a supply section **92** and a discharge section **93**. The supply section **92** and the discharge section **93** are attached to the respective sides of the housing **91** in the second direction **y**. The supply section **92** supplies the cooling medium **Cm** to the housing **91**. The discharge section **93** discharges the cooling medium **Cm** that has flowed through the housing **91**. In this way, the cooling medium **Cm** flows in the second direction **y** in the housing **91**. Note that “the cooling medium **Cm** flows in the second direction **y**” does not mean that only the flow velocity component in the second direction **y** exists, but includes the state in which the cooling medium **Cm** moves in the second direction **y** as a whole while including the flow velocity components in the first direction **x** and the thickness direction **z**.

[0118] Next, the advantages of the present embodiment will be described.

[0119] The first heat dissipation member **2A** has the first bases **21A** and the first upright portions **22A**. The first upright portions **22A** protrude from the first reverse surface **302A** in the thickness direction **z** to increase the heat transfer area. Each of the first bases **21A** is housed in a first recess **303A** and bonded to the first reverse-surface metal layer **33A** within the first recess **303A**. This makes it possible to more accurately position the first heat dissipation member **2A** to the first substrate **3A** when attaching the first heat dissipation member **2A** to the first substrate **3A** after forming the first substrate **3A**, the first semiconductor elements **10A**, the second semiconductor elements **10B**, and the sealing resin **8**. This can reduce the labor required for manufacturing. Thus, it is possible to improve heat dissipation efficiency and save labor during the manufacturing process.

[0120] The first bases **21A** and the first reverse-surface metal layer **33A**, which are made of different metals, are joined by laser bonding, whereby the heat generated during the joining process can be reduced. This reduces the possibilities of unintended damage or the like to the first semiconductor elements **10A**, the second semiconductor elements **10B**, and the conductive paths of these semiconductor elements.

[0121] The first heat dissipation member **2A** includes the first connecting portions **23A**. As such, the first heat dissipation member **2A** is configured with a plurality of water channels each having a rectangular shape as viewed in the second direction **y**. This further improves the heat dissipation efficiency of the electric power conversion unit **B1**.

[0122] The first bases **21A**, the first upright portions **22A**, and the first connecting portions **23A** extend in the second direction **y**. As such, a plurality of flow channels are formed to extend across the first reverse surface **302A** in the second direction **y**. The supply section **92** and the discharge section **93** of the cooling device **9** are disposed on the respective sides in the second direction **y**.

This allows the cooling medium C_m to flow along the first heat dissipation member **2A** in the second direction y , thus improving the heat dissipation efficiency of the semiconductor device **A1**. [0123] FIGS. **25** to **28** show other embodiments of the present disclosure. In these figures, elements that are the same as or similar to those in the above embodiment are provided with the same reference numerals as in the above embodiment. The configurations of the elements in each embodiment can be combined as appropriate as long as the combination does not cause technical inconsistency.

Second Embodiment

[0124] FIGS. **25** to **27** show a heat dissipation member of a semiconductor device according to a second embodiment of the present disclosure. A semiconductor device **A2** of the present embodiment is different from the above embodiment in that the semiconductor device **A2** includes a plurality of first heat dissipation members **2A** and a plurality of second heat dissipation members **2B**, and in the configuration of the first substrate **3A**.

[0125] As shown in FIGS. **26** and **27**, the first reverse-surface metal layer **33A** of the present embodiment includes a plurality of first recesses **303A** and a plurality of second recesses **303B**.

[0126] The first recesses **303A** and the second recesses **303B** are recessed from the first reverse surface **302A** to the z_1 side in the thickness direction z . In the present embodiment, the first recesses **303A** are aligned in the first direction x . The second recesses **303B** are also aligned in the first direction x . The first recesses **303A** and the second recesses **303B** are alternately arranged in the second direction y . A first recess **303A** and a second recess **303B** adjacent in the second direction y are shifted in position in the first direction x . Although the first recesses **303A** and the second recesses **303B** of the present embodiment are rectangular as viewed in the thickness direction z , the shape of each of these recesses is not particularly limited.

[0127] The first heat dissipation members **2A** and the second heat dissipation members **2B** are disposed on the surface of the first substrate **3A** (the first reverse-surface metal layer **33A**) on the z_2 side in the thickness direction z . The first heat dissipation members **2A** and the second heat dissipation members **2B** are alternately arranged in the second direction y . Each of the first heat dissipation members **2A** may have the same configuration as the first heat dissipation member **2A** in the semiconductor device **A1** except that the first heat dissipation member **2A** in the semiconductor device **A2** is smaller in the second direction y than the first heat dissipation member **2A** in the semiconductor device **A1**.

[0128] Each of the second heat dissipation members **2B** has a plurality of second bases **21B**, a plurality of second upright portions **22B**, and a plurality of second connecting portions **23B**. The material of the second heat dissipation members **2B** is not particularly limited. For example, each of the second heat dissipation members **2B** is made of a metal plate material. The metal plate material may contain a metal such as copper (Cu), aluminum (Al), or stainless steel, or an alloy of these metals.

[0129] The second bases **21B** are located on the z_1 side in the thickness direction z . The second bases **21B** are housed in the respective second recesses **303B**. In the second recesses **303B**, the second bases **21B** are bonded to the first reverse-surface metal layer **33A**. The method for bonding the second bases **21B** to the first reverse-surface metal layer **33A** is not particularly limited, and can be selected as appropriate from a welding method such as laser welding, a method using joints such as brazing, or other methods such as ultrasonic bonding or solid-phase diffusion bonding. In the illustrated example, the second bases **21B** are bonded to the first reverse-surface metal layer **33A** by laser bonding. In this case, parts of the second bases **21B** and parts of the first reverse-surface metal layer **33A** are melted and solidified to form welded portions **M**. In the present embodiment, the thickness of each second base **21B** in the thickness direction z is smaller than the depth of each second recess **303B** in the thickness direction z . The shape of each second base **21B** is not particularly limited. In the present embodiment, each second base **21B** has a strip shape extending in the second direction y . Each second base **21B** may be sized and shaped to fit in a second recess

303B, or may be slightly smaller than a second recess **303B**.

[0130] The second upright portions **22B** are connected to the respective ends of the second bases **21B** in the first direction *x*, and stand in the thickness direction *z*. The second upright portions **22B** protrude beyond the first reverse surface **302A** to the *z2* side in the thickness direction *z*. In the present embodiment, the length of each second upright portion **22B** in the second direction *y* is equal (or substantially equal) to the length of each second base **21B** in the second direction *y*. Each of the second upright portions **22B** has a strip shape extending in the second direction *y*, for example.

[0131] Each of the second connecting portions **23B** connects the *z2*-side ends in the thickness direction *z* of second upright portions **22B** adjacent to each other in the first direction *x*. In the present embodiment, the length of each second connecting portion **23B** in the second direction *y* is equal (or substantially equal) to the length of each second upright portion **22B** in the second direction *y*. The shape of each second connecting portion **23B** as viewed in the thickness direction *z* is not particularly limited, and may be a strip shape in the present embodiment. The shape of each second connecting portion **23B** as viewed in the second direction *y* is not particularly limited, and may be a flat shape along the first direction *x* as illustrated, or may be a dome shape or a ridge shape that bulges to the *z2* side in the thickness direction *z*.

[0132] Since a first recess **303A** and a second recess **303B** adjacent in the second direction *y* are shifted in position in the first direction *x*, a first connecting portion **23A** and a second connecting portion **23B** adjacent in the second direction *y* are also shifted in position in the first direction *x*.

[0133] The present embodiment can also improve heat dissipation efficiency and save labor during the manufacturing process. The flow channels formed by the first heat dissipation members **2A** are offset from the flow channels formed by the second heat dissipation members **2B** in the first direction *x*. Thus, the cooling medium *Cm* is likely to meander when flowing from the supply section **92** to the discharge section **93** in the cooling device **9**. This facilitates heat transfer from the first heat dissipation members **2A** and the second heat dissipation members **2B**, thereby further improving heat dissipation efficiency.

[0134] As can be understood from the present embodiment, the respective numbers of first heat dissipation members **2A** and second heat dissipation members **2B** are not particularly limited.

Third Embodiment

[0135] FIG. **28** shows a semiconductor device according to a third embodiment of the present disclosure. A semiconductor device **A3** of the present embodiment is different from the above embodiments in that the semiconductor device **A3** includes a first heat dissipation member **2A**, a third heat dissipation member **2C**, a first substrate **3A**, and a second substrate **3B**.

[0136] The second substrate **3B** is disposed on the *z1* side in the thickness direction *z* with respect to the first semiconductor elements **10A** and the second semiconductor elements **10B**. The specific configuration of the second substrate **3B** is not particularly limited. For example, the second substrate **3B** may be a direct bonded copper (DBC) substrate or an active metal brazing (AMB) substrate. The second substrate **3B** may be directly bonded to at least either of the first semiconductor elements **10A** and the second semiconductor elements **10B**, or may be configured to transfer heat via a bonding layer (not illustrated) or a metal member (not illustrated).

[0137] In the illustrated example, the second substrate **3B** has a second reverse-surface metal layer **33B**. The second reverse-surface metal layer **33B** is made of the same material as the first reverse-surface metal layer **33A** described above. The second reverse-surface metal layer **33B** has a second reverse surface **302B** and a plurality of third recesses **303C**.

[0138] The second reverse surface **302B** faces the *z1* side in the thickness direction *z*. The second reverse surface **302B** is exposed from the sealing resin **8**. The third recesses **303C** are recessed from the second reverse surface **302B** to the *z2* side in the thickness direction *z*. In the present embodiment, the third recesses **303C** are aligned in the first direction *x*. The specific shape of each third recess **303C** is not particularly limited. For example, each third recess **303C** may be a groove

extending in the second direction y or may have a rectangular shape as viewed in the thickness direction z.

[0139] The third heat dissipation member **2C** is disposed on the surface of the second substrate **3B** (the second reverse-surface metal layer **33B**) on the z1 side in the thickness direction z. The third heat dissipation member **2C** has a plurality of third bases **21C**, a plurality of third upright portions **22C**, and a plurality of third connecting portions **23C**. The material of the third heat dissipation member **2C** is not particularly limited. For example, the third heat dissipation member **2C** is made of a metal plate material. The metal plate material may contain a metal such as copper (Cu), aluminum (Al), or stainless steel, or an alloy of these metals.

[0140] The third bases **21C** are located on the z2 side in the thickness direction z. The third bases **21C** are housed in the respective third recesses **303C**. In the third recesses **303C**, the third bases **21C** are bonded to the second reverse-surface metal layer **33B**. The method for bonding the third bases **21C** to the second reverse-surface metal layer **33B** is not particularly limited, and can be selected as appropriate from a welding method such as laser welding, a method using joints such as brazing, or other methods such as ultrasonic bonding or solid-phase diffusion bonding. In the present embodiment, the thickness of each third base **21C** in the thickness direction z is smaller than the depth of each third recess **303C** in the thickness direction z. The shape of each third base **21C** is not particularly limited. For example, each third base **21C** has a strip shape extending in the second direction y. Each third base **21C** may be sized and shaped to fit in a third recess **303C**, or may be slightly smaller than a third recess **303C**.

[0141] The third upright portions **22C** are connected to the respective ends of the third bases **21C** in the first direction x, and stand in the thickness direction z. The third upright portions **22C** protrude beyond the second reverse surface **302B** to the z1 side in the thickness direction z. In the present embodiment, the length of each third upright portion **22C** in the second direction y is equal (or substantially equal) to the length of each third base **21C** in the second direction y. The shape of each third upright portion **22C** is not particularly limited. For example, each third upright portion **22C** has a strip shape extending in the second direction y.

[0142] Each of the third connecting portions **23C** connects the z1-side ends in the thickness direction z of third upright portions **22C** adjacent to each other in the first direction x. In the present embodiment, the length of each third connecting portion **23C** in the second direction y is equal (or substantially equal) to the length of each third upright portion **22C** in the second direction y. The shape of each third connecting portion **23C** as viewed in the thickness direction z is not particularly limited, and may be a strip shape in the present embodiment. The shape of each third connecting portion **23C** as viewed in the second direction y is not particularly limited, and may be a flat shape along the first direction x as illustrated, or may be a dome shape or a ridge shape that bulges to the z1 side in the thickness direction z.

[0143] The present embodiment can also improve heat dissipation efficiency and save labor during the manufacturing process. In addition, the first heat dissipation member **2A** and the third heat dissipation member **2C** are disposed on the opposite sides in the thickness direction z to cool the semiconductor device **A3** from both sides in the thickness direction z. This is advantageous for improving heat dissipation efficiency.

[0144] The semiconductor device, the electric power conversion unit, and the method for manufacturing the semiconductor device according to the present disclosure are not limited to the above embodiments. Various design changes can be made to the specific configurations of the semiconductor device, the electric power conversion unit, and the method for manufacturing the semiconductor device according to the present disclosure. The present disclosure includes the embodiments described in the following clauses.

[0145] Clause 1. A semiconductor device comprising: [0146] a semiconductor element; [0147] a first substrate supporting the semiconductor element; [0148] a sealing resin covering the semiconductor element and a part of the first substrate; and [0149] a first heat dissipation member,

[0150] wherein the first substrate includes an obverse surface facing a first side in a thickness direction, and a first reverse surface facing a second side in the thickness direction and exposed from the sealing resin, [0151] the semiconductor element is mounted on the obverse surface, [0152] the first heat dissipation member is disposed on a surface of the first substrate on the second side in the thickness direction, [0153] the first heat dissipation member includes a plurality of first bases located on the first side in the thickness direction, and a plurality of first upright portions extending from the plurality of first bases to the second side in the thickness direction, [0154] the first substrate includes a plurality of first recesses recessed from the first reverse surface to the first side in the thickness direction, and [0155] the plurality of first bases are housed in the respective first recesses.

[0156] Clause 2. The semiconductor device according to clause 1, wherein the first substrate includes a first reverse-surface metal layer formed with the plurality of first recesses, and [0157] the first heat dissipation member is bonded to the first reverse-surface metal layer.

[0158] Clause 3. The semiconductor device according to clause 2, wherein the first heat dissipation member is made of metal.

[0159] Clause 4. The semiconductor device according to clause 2 or 3, wherein the plurality of first recesses are aligned in a first direction perpendicular to the thickness direction.

[0160] Clause 5. The semiconductor device according to clause 4, wherein the plurality of first recesses are a plurality of grooves extending in a second direction perpendicular to the thickness direction and the first direction.

[0161] Clause 6. The semiconductor device according to clause 5, wherein the first heat dissipation member includes a first connecting portion that connects second-side ends of two first upright portions adjacent in the first direction out of the plurality of first upright portions, the second-side ends being ends of the adjacent first upright portions located on the second side in the thickness direction.

[0162] Clause 7. The semiconductor device according to clause 4, further comprising a second heat dissipation member disposed on the surface of the first substrate on the second side in the thickness direction, [0163] wherein the second heat dissipation member includes a plurality of second bases located on the first side in the thickness direction, and a plurality of second upright portions extending from the second bases to the second side in the thickness direction, [0164] the first substrate includes a plurality of second recesses recessed from the first reverse surface to the first side in the thickness direction, [0165] the plurality of second bases are housed in the respective second recesses, and [0166] the plurality of first recesses and the plurality of second recesses are adjacent to each other in a second direction perpendicular to the thickness direction and the first direction.

[0167] Clause 8. The semiconductor device according to clause 7, wherein the second heat dissipation member is bonded to the first reverse-surface metal layer.

[0168] Clause 9. The semiconductor device according to clause 8, wherein the second heat dissipation member is made of metal.

[0169] Clause 10. The semiconductor device according to clause 8 or 9, wherein the plurality of second recesses are aligned in the first direction.

[0170] Clause 11. The semiconductor device according to any of clauses 7 to 10, wherein the plurality of first recesses include a first recess that is adjacent to one of the plurality of second recesses in the second direction, and the first recess and the second recess, which are adjacent to each other in the second direction, are shifted in position in the first direction.

[0171] Clause 12. The semiconductor device according to any of clauses 2 to 11, wherein the first substrate includes a first insulating layer located on the first side in the thickness direction with respect to the first reverse-surface metal layer, and a first metal layer located on the first side in the thickness direction with respect to the first insulating layer.

[0172] Clause 13. The semiconductor device according to any of clauses 1 to 12, further

comprising: [0173] a second substrate located on the first side in the thickness direction with respect to the semiconductor element; and [0174] a third heat dissipation member, [0175] wherein the second substrate includes a second reverse surface facing the first side in the thickness direction and exposed from the sealing resin, [0176] the third heat dissipation member is disposed on a surface of the second substrate on the first side in the thickness direction, [0177] the third heat dissipation member includes a plurality of third bases located on the second side in the thickness direction, and a plurality of third upright portions extending from the third bases to the first side in the thickness direction, [0178] the second substrate includes a plurality of third recesses recessed from the second reverse surface to the second side in the thickness direction, and [0179] the plurality of third bases are housed in the respective third recesses.

[0180] Clause 14. The semiconductor device according to clause 13, wherein the second substrate includes a second reverse-surface metal layer formed with the plurality of third recesses, and [0181] the third heat dissipation member is bonded to the second reverse-surface metal layer.

[0182] Clause 15. The semiconductor device according to clause 14, wherein the third heat dissipation member is made of metal.

[0183] Clause 16. The semiconductor device according to clause 14 or 15, wherein the plurality of third recesses are aligned in the first direction perpendicular to the thickness direction.

[0184] Clause 17. An electric power conversion unit comprising: [0185] the semiconductor device according to any of clauses 1 to 16, and [0186] a cooling device disposed on the second side in the thickness direction with respect to the semiconductor device, and [0187] wherein the cooling device includes a housing that houses the first heat dissipation member, and that allows a cooling medium to flow.

REFERENCE NUMERALS

TABLE-US-00001 A1, A2, A3: Semiconductor device B1: Electric power conversion unit 2A: First heat dissipation member 2B: Second heat dissipation member 2C: Third heat dissipation member 3: Support substrate 3A: First substrate 3B: Second substrate 5: First conductive member 6: Second conductive member 8: Sealing resin 9: Cooling device 10A: First semiconductor element 10B: Second semiconductor element 11: First obverse-surface electrode 12: Second obverse-surface electrode 13: Third obverse-surface electrode 15: Reverse-surface electrode 17: Thermistor 19A: First conductive bonding member 19B: Second conductive bonding member 21A: First base 21B: Second base 21C: Third base 22A: First upright portion 22B: Second upright portion 22C: Third upright portion 23A: First connecting portion 23B: Second connecting portion 31A: First insulating layer 32A: First obverse-surface metal layer 33A: First reverse-surface metal layer 33B: second reverse-surface metal layer 41: First terminal 42: Second terminal 43: Third terminal 44: Fourth terminal 45: Control terminal 46A-46E: First control terminal 47A-47D: Second control terminal 48: Control terminal support 48A: First support portion 48B: Second support portion 49: Bonding member 51: Main portion 52: First bonding portion 53: Second bonding portion 59: Conductive bonding member 61: Third bonding portion 64: First path portion 65: Second path portion 66: Third path portion 67: Fourth path portion 69: Conductive bonding member 71-74: Wire 81: Resin obverse surface 82: Resin reverse surface 86: Resin cavity 89: Groove 91: Housing 92: Supply section 93: Discharge section 101: Element obverse surface 102: Element reverse surface 121: Gate finger 301A: First obverse surface 301B: Second obverse surface 302A: First reverse surface 302B: Second reverse surface 303A: First recess 303B: Second recess 303C: Third recess 321: First conductive portion 322: Second conductive portion 451: Holder 452: Metal pin 459: Conductive bonding member 481: Insulating layer 482: First metal layer 482A: First region 482B: Second region 482C: Third region 482D: Fourth region 482E: Fifth region 482F: Sixth region 483: Second metal layer 514: First opening 602: First ramp portion 603: Second ramp portion 611: Flat section 612: First inclined section 641: First band-shaped section 643: First extended section 649: Recess 651: Second band-shaped section 653: Second extended section 659, 669: Recess 831-834: Resin side surface 832a: Recess 851: First protrusion 851a: First protrusion

end surface 851b: Recess 851c: Inner wall surface 852: Second protrusion 911: Groove 919:
Sealant Cm: Cooling medium M: Welded portion x: First direction y: Second direction z: Thickness
direction

Claims

1. A semiconductor device comprising: a semiconductor element; a first substrate supporting the semiconductor element; a sealing resin covering the semiconductor element and a part of the first substrate; and a first heat dissipation member, wherein the first substrate includes an obverse surface facing a first side in a thickness direction, and a first reverse surface facing a second side in the thickness direction and exposed from the sealing resin, the semiconductor element is mounted on the obverse surface, the first heat dissipation member is disposed on a surface of the first substrate on the second side in the thickness direction, the first heat dissipation member includes a plurality of first bases located on the first side in the thickness direction, and a plurality of first upright portions extending from the plurality of first bases to the second side in the thickness direction, the first substrate includes a plurality of first recesses recessed from the first reverse surface to the first side in the thickness direction, and the plurality of first bases are housed in the respective first recesses.
2. The semiconductor device according to claim 1, wherein the first substrate includes a first reverse-surface metal layer formed with the plurality of first recesses, and the first heat dissipation member is bonded to the first reverse-surface metal layer.
3. The semiconductor device according to claim 2, wherein the first heat dissipation member is made of metal.
4. The semiconductor device according to claim 2, wherein the plurality of first recesses are aligned in a first direction perpendicular to the thickness direction.
5. The semiconductor device according to claim 4, wherein the plurality of first recesses are a plurality of grooves extending in a second direction perpendicular to the thickness direction and the first direction.
6. The semiconductor device according to claim 5, wherein the first heat dissipation member includes a first connecting portion that connects second-side ends of two first upright portions adjacent in the first direction out of the plurality of first upright portions, the second-side ends being ends of the adjacent first upright portions located on the second side in the thickness direction.
7. The semiconductor device according to claim 4, further comprising a second heat dissipation member disposed on a surface of the first substrate on the second side in the thickness direction, wherein the second heat dissipation member includes a plurality of second bases located on the first side in the thickness direction, and a plurality of second upright portions extending from the second bases to the second side in the thickness direction, the first substrate includes a plurality of second recesses recessed from the first reverse surface to the first side in the thickness direction, the plurality of second bases are housed in the respective second recesses, and the plurality of first recesses and the plurality of second recesses are adjacent to each other in a second direction perpendicular to the thickness direction and the first direction.
8. The semiconductor device according to claim 7, wherein the second heat dissipation member is bonded to the first reverse-surface metal layer.
9. The semiconductor device according to claim 8, wherein the second heat dissipation member is made of metal.
10. The semiconductor device according to claim 8, wherein the plurality of second recesses are aligned in the first direction.
11. The semiconductor device according to claim 7, wherein the plurality of first recesses include a first recess that is adjacent to one of the plurality of second recesses in the second direction, and the

first recess and the second recess, which are adjacent to each other in the second direction, are shifted in position in the first direction.

12. The semiconductor device according to claim 2, wherein the first substrate includes a first insulating layer located on the first side in the thickness direction with respect to the first reverse-surface metal layer, and a first metal layer located on the first side in the thickness direction with respect to the first insulating layer.

13. The semiconductor device according to claim 1, further comprising: a second substrate located on the first side in the thickness direction with respect to the semiconductor element; and a third heat dissipation member, wherein the second substrate includes a second reverse surface facing the first side in the thickness direction and exposed from the sealing resin, the third heat dissipation member is disposed on a surface of the second substrate on the first side in the thickness direction, the third heat dissipation member includes a plurality of third bases located on the second side in the thickness direction, and a plurality of third upright portions extending from the third bases to the first side in the thickness direction, the second substrate includes a plurality of third recesses recessed from the second reverse surface to the second side in the thickness direction, and the plurality of third bases are housed in the respective third recesses.

14. The semiconductor device according to claim 13, wherein the second substrate includes a second reverse-surface metal layer formed with the plurality of third recesses, and the third heat dissipation member is bonded to the second reverse-surface metal layer.

15. The semiconductor device according to claim 14, wherein the third heat dissipation member is made of metal.

16. The semiconductor device according to claim 14, wherein the plurality of third recesses are aligned in the first direction perpendicular to the thickness direction.

17. An electric power conversion unit comprising: the semiconductor device according to claim 1, and a cooling device disposed on the second side in the thickness direction with respect to the semiconductor device, and wherein the cooling device includes a housing that houses the first heat dissipation member, and that allows a cooling medium to flow.
